

Historical ciphers: ciphertext-only solutions

cyphersolver project (Daniel Bourdeau, with George Lasry) · 22 ciphertext-only first breaks and 3 further Cipherbrain challenges · list of 25 September 2026 · <https://dbourdeau.github.io/cyphersolver/>

Key recovered from ciphertext only

First breaks: the key was rebuilt from the ciphertext alone, with no plaintext and no existing key. Items that someone had solved earlier but never published are included and marked.

1. Sienese envoys and others to the government of Siena before 1429 to 1547

Keys recovered from the ciphertext alone for three letters (nos. 25, 18, 14) and from an alphabet fragment for a fourth; none of the 43 keys filed with the letters fits. No. 25 reads 96.2%, no. 18 (Latin treaty articles with the Emperor) 87.1%. Several other letters in the file remain open.

Method	ciphertext-only break · partial
Primary source	Archivio di Stato di Siena, Concistoro 2308, fasc. 2 "Lettere in cifra"
Secondary source	DECODE R4790-R4814 (no status); the archive's 1952 inventory calls the letters 'solo parzialmente decifrate'
Write-up	https://dbourdeau.github.io/cyphersolver/siena1421.html

2. Matheo de Segura to the Constable of Castile Madrid, 8 Aug 1596

Two short passages in a homophonic cipher; the key was rebuilt by annealing and resegmenting the image. 119 of 125 cipher tokens read (95.2%). The passages concern predictions of the world ending or the king dying by September.

Method	ciphertext-only break · complete
Primary source	BnF, espagnol 336 no. 99, ff. 196r-198r
Secondary source	Tomokiyo, cryptiana (spanish3.htm, unsolved.htm): the last es. 336 item listed as undeciphered
Write-up	https://dbourdeau.github.io/cyphersolver/segura1596.html

3. Giovan Gioacchino da Passano (attributed), London, to Francis I 15 Mar 1530 (copy heading)

An account of Wolsey's wealth: cash, credits, revenues, jewels and plate, Winchester, St Albans and York. A dotted re-transcription made the break possible. 386 of 402 signs read coherently (96.0%).

Method	ciphertext-only break · complete
Primary source	BnF, Clairambault 331, ff. 156-157
Secondary source	Tomokiyo, cryptiana (francis.htm), listed as undeciphered
Write-up	https://dbourdeau.github.io/cyphersolver/joachim1530.html

4. Mémoire en chiffre to the connétable de Montmorency 12 Dec 1560

Ten pages in a cursive pseudo-script: a homophonic substitution with two-stroke units and four word codes, broken from the images alone. It reports on the Protestant rising in the Agenais a week after François II's death. 98.9% of 1,715 cipher words read.

Method	ciphertext-only break · complete
Primary source	BnF, français 3157 no. 67, ff. 152r-156v
Secondary source	BnF catalogue (Gallica harvest); no DECODE record
Earlier solution	In private communications, Lasry wrote that he independently solved it in 2023 (improved in February 2026) but his solution has not been published.
Write-up	https://dbourdeau.github.io/cyphersolver/memoire1560.html

5. "L'Estat du Roy de Navarre et de son party en France" after 1580

A French memoir on Navarre's party in a homophonic alphabet of about 50 graphic signs. The letter m is written as a group of dots, which earlier passes took for punctuation. 95.1% read.

Method	ciphertext-only break · complete
Primary source	British Library, Harley MS 1582, ff. 263-264
Secondary source	DECODE R8505 (Non-decrypte, "unknown sender")
Write-up	https://dbourdeau.github.io/cyphersolver/harley1582r8505.html

6. Ferenc Wesselényi, Palatine of Hungary, to Leopold I c. 1663-64

A reversed-alphabet cipher with vowel homophones in a Latin draft, broken by annealing and then corrected by hand. A general is to take His Majesty's garrison by force if need be. 98.8% of cipher tokens read; two codes are open.

Method	ciphertext-only break · complete
Primary source	MNL OL E 199 (Wesselényi family archive), 8. pallium 1. 01
Secondary source	DECODE R674 (non-decrypte)
Write-up	https://dbourdeau.github.io/cyphersolver/wesselenyi1664.html

7. Sir Simonds D'Ewes, cipher log of his son's convulsion fits 3 Apr 1635 - Jan 1636

D'Ewes's private alphabet over Latin, rebuilt from cribs (convulsio, plena luna); the weekdays and moon phases check out against his Autobiography. 96.5% of 287 words read.

Method	ciphertext-only break · complete
Primary source	British Library, Harley MS 286, f. 61
Secondary source	DECODE R7759 ("unknown sender to unknown recipient, 1635")
Write-up	https://dbourdeau.github.io/cyphersolver/harley286.html

8. Unknown writer to Hesse-Kassel undated (volume 1635-52)

A digit-and-sign letter substitution. None of the volume's ~120 key sheets fits; the key was annealed on German of the period and confirmed by the contemporary gloss on lines 1-4. About 97% read.

Method	ciphertext-only break · complete
Primary source	Hessisches Staatsarchiv Marburg, HStAM 4 d Nr. 1218, no. 49
Secondary source	DECODE R4500 (partially decrypte)
Write-up	https://dbourdeau.github.io/cyphersolver/marburg1635.html

9. Ferdinand of Bavaria, Elector of Cologne, to Duke Maximilian I of Bavaria Bonn, 22 Dec 1619

Ten pages in a homophonic substitution on two-digit figures with a nomenclator, broken by annealing 2,482 letters; the key reads the sibling R9425 unchanged. The letter covers quartering and troops six weeks after Frederick V's coronation. 95.8% read.

Method	ciphertext-only break · complete
Primary source	Bayerisches Hauptstaatsarchiv, Kurbayern Äußeres Archiv 4591, ff. 281-286
Secondary source	DECODE R9426 (non-decrypte, no author or recipient)
Write-up	https://dbourdeau.github.io/cyphersolver/ferdinand1619.html

10. Swatow telegram to Sun Yat-sen 3 Apr 1916

Standard Chinese telegraph code passed through a systematic code condenser, recovered by brute force. 41 of about 44 characters read.

Method	ciphertext-only break · complete
Primary source	JACAR (Japan Center for Asian Historical Records) B03050738800
Secondary source	Tomokiyo, cryptiana list and blog (12 Dec 2021), unsolved since 2021
Write-up	https://dbourdeau.github.io/cyphersolver/sunyatsen.html

11. Sun Yat-sen's intercepted telegrams Mar 1916 - Mar 1917

About 60 condenser telegrams read under seven keys of the Swatow family, plus the Hankow +111 digit code, including Chen Qimei's assassination as reported to Tokyo. Sun's six-vowel 文密 code is still open.

Method	ciphertext-only break · complete
Primary source	JACAR, 特定人 / 発受郵便物取調 (Ministry of Communications intercepts)
Secondary source	Follow-on from the Swatow telegram (cryptiana)
Write-up	https://dbourdeau.github.io/cyphersolver/sunintercepts.html

12. Warsaw letter to the Bishop of Olmütz (?) 24 Dec 1627

A homophonic substitution whose alphabet is in plain order, with syllables, nulls and thirteen word codes, recovered by annealing. The writer presses for a canonry of Olmütz for a son of the Queen of Poland. Read in full; the word codes are glossed from context.

Method	ciphertext-only break · complete
Primary source	ÖStA HHStA, Staatskanzlei Interiora, Chiffrenschlüssel Kt. 14 Fasc. 20 f. 176
Secondary source	DECODE R1408 (non-decrypting); transcription by Tomokiyo, cryptiana (variable2.htm, unsolved.htm)
Write-up	https://dbourdeau.github.io/cyphersolver/warsaw.html

13. Henry of Navarre to Ségur 1585-86

Three figure-cipher letters to the envoy raising a German army. The key, an alphabetical syllabary, was rebuilt by a structured annealer and checked on matched controls. The letters cover Duke Casimir, Clervant's reiters and the road through the Vivarais.

Method	ciphertext-only break · complete
Primary source	BnF, 500 de Colbert 401, ff. 233, 239, 288v
Secondary source	Tomokiyo, cryptiana unsolved list (as letters of Henry III)
Earlier solution	In private communications, Lasry wrote that he independently solved it in 2024 but his solution has not been published.
Write-up	https://dbourdeau.github.io/cyphersolver/segur.html

14. Cardinal Soglia to nuncio Viale Prelà at Innsbruck Rome, 15 June 1848

A 64-cell two-digit table with homophones and syllables, 5 as word separator, and an alphabetical one-part code. It is a counter-order: stay with the Emperor and suspend the request for passports. Ten code words read from context; 11 code groups open.

Method	ciphertext-only break · complete
Primary source	L'Italia del Popolo no. 40 (Milan, 30 June 1848), Roman broadsheet reprint
Secondary source	Cipherbrain challenge (Klaus Schmeh, 13 Dec 2014)
Cipherbrain	also a Cipherbrain challenge
Write-up	https://dbourdeau.github.io/cyphersolver/soglia1848.html

15. Gottfried van Swieten to Count Cobenzl Bonn, 1757-59

The 1759 table (groups 100-442) was rebuilt ciphertext-only by monotone annealing against French, then corrected by hand. The despatches cover the Gueldre convention and Soubise's army. 96% of 553 groups read.

Method	ciphertext-only break · complete
Primary source	Archives générales du Royaume, Brussels, Secrétairerie d'État et de Guerre 1236
Secondary source	DECODE R955-R957 (non-decrypting)
Write-up	https://dbourdeau.github.io/cyphersolver/swieten1757.html

16. Johann Heinrich Kauderbach (The Hague) to Brühl / Friedrich August II 1754-56

Unseparated figures in which 5 and 8 are nulls; one key serves all ten despatches. It was found by one-to-one annealing onto the vocabulary of his 1761 key. All ten despatches read at 65-97%, covering the Dutch augmentation debate and the Treaty of Versailles.

Method ciphertext-only break · complete
Primary source Koninklijk Huisarchief, The Hague, Prins Willem V inv. 201
Secondary source DECODE R1043, R1044, R1954, R1958, R1959, R2078-R2082 ("never solved" by the Hague cabinet)
Write-up <https://dbourdeau.github.io/cyphersolver/kauderbach1754.html>

17. Potocka and Mniszech to Jakub Dunin 1714-16

Potocka's alphabet was broken from a Polish postscript and carried over to eight French letters; 97.3% of Potocka tokens valued. Mniszech's letter is still unread.

Method ciphertext-only break · partial
Primary source Archiwum Narodowe w Krakowie, Archiwum Sanguszków
Secondary source DECODE R7524-R7530, R7534-R7536
Write-up <https://dbourdeau.github.io/cyphersolver/potocka1714.html>

18. Four cipher letters to Gianfrancesco Gonzaga, lord of Mantua 1428-1430

Two keys rebuilt ciphertext-only: Rome, 9 Aug 1428 (96.6%) and Cattanei, Feb 1430 (96.7%). A third letter was re-deciphered from its interlinear gloss. Pandolfo Malatesta's own runs remain open. 91.5% of all signs read.

Method ciphertext-only break · partial
Primary source Archivio di Stato di Mantova, Archivio Gonzaga
Secondary source DECODE R1854 (partially decrypted; catalogued as one letter)
Write-up <https://dbourdeau.github.io/cyphersolver/r1854.html>

19. Guido Rangone (Venice) to Anne de Montmorency 24 Dec 1529 and 10 Jan 1530

A homophonic substitution of about 50 signs for Italian, broken by annealing. Both passages concern Cesare Fregoso. 87.9% of signs read; four short runs are too short to resolve.

Method ciphertext-only break · partial
Primary source BnF, français 3070 f. 81 no. 40 and français 3082 f. 42 no. 19
Secondary source DECODE R4249; Tomokiyo, cryptiana (francis.htm) lists both as undeciphered
Write-up <https://dbourdeau.github.io/cyphersolver/rangone1530.html>

20. Thomas Edmondes (at Henri IV's court) to Lord Treasurer Burghley Jan 1592 - Jul 1594

Six letters from the King's camp with lines in invented signs: a simple substitution of English with homophones and person signs, broken from the clear context. Everything reads except four lines on f. 59.

Method ciphertext-only break · partial
Primary source British Library, Stowe MS 166, ff. 50-61, 109-112
Secondary source DECODE R7770-R7775 ("Lord Threr? to unknown recipient")
Write-up <https://dbourdeau.github.io/cyphersolver/stowe166.html>

21. Claude de La Guiche, bishop of Mirepoix (Rome), to Montmorency 22 Nov 1551

A 406-sign passage in 22 invented signs, broken by permutation annealing. It covers the Count of Santa Fiora, 'mal satisfait' and ready to serve the king on conditions. Seure's Lisbon letters in the same file are left open.

Method ciphertext-only break · partial
Primary source BnF, français 3138 no. 22, f. 60v
Secondary source BnF catalogue (Gallica harvest)
Write-up <https://dbourdeau.github.io/cyphersolver/guiche1551.html>

22. An Orange prince to an officer of the Dutch émigré troops c. 1795

Vertical digit pairs from a 6x6 square, recovered by quadgram annealing on the Dutch postscript; the square then read the French page unchanged. The letter concerns General Dundas, pay and embarkation. 94% of cipher tokens read; about thirty word signs open.

Method	ciphertext-only break · partial
Primary source	Koninklijk Huisarchief, The Hague, Prins Willem V inv. 339
Secondary source	DECODE R1892 (non-decrypted)
Earlier solution	In private communications, Lasry wrote that he independently solved it in 2021 but his solution has not been published.
Write-up	https://dbourdeau.github.io/cyphersolver/r1892.html

Other Cipherbrain challenges (any method)

Challenges posted on Klaus Schmeh's Cipherbrain that this project read by other routes. The Soglia dispatch above is also a Cipherbrain challenge.

23. Johan Banér to Torsten Stålhandske (intercepted copy) Hof, 29 Dec 1640

A homophonic two-digit alphabet with 14 code words; none of the Swedish keys on DECODE fits. The key was rebuilt from context cribs (durch, und): the letter orders a march toward the Upper Palatinate. 89.8% of tokens read; the code words are open.

Method	key from adjacent plaintext · partial
Primary source	Archivio della Pontificia Università Gregoriana, APUG 568 f. 239 (Kircher correspondence)
Secondary source	Cipherbrain challenge (Klaus Schmeh, 2 June 2019)
Write-up	https://dbourdeau.github.io/cyphersolver/baner1640.html

24. Pseudo-Arndt, "Das zweyte Silentium Dei" Glücksbrunn, 1798

Read with the key drawn in the book itself (p. 83), so no key was recovered. All 31 cipher passages (1,545 signs) were transcribed and read. Page 24 had already been read by Lasry and Kopal (Cipherbrain 2022); five short passages are read here for the first time and earlier readings corrected.

Method	known key applied · complete
Primary source	Beinecke Library, Mellon MS 136, pp. 24-76
Secondary source	Cipherbrain (Klaus Schmeh, 4 March 2022); DECODE R2873
Write-up	https://dbourdeau.github.io/cyphersolver/mellon136.html

25. Thomas Urquhart's Cyphral Octastich 1652

A book cipher on the book itself, verified independently: 238 of 284 hits against shuffled controls, reading a royalist prayer for Charles II. The same rule and plaintext had been published by Vals AI in Aug 2026, so this is a verification, not a first break. The distich claim was not reproduced.

Method	known key applied · partial
Primary source	Last leaf of Ekskybalaaron / The Jewel (London, 1652); Jack and Lyall edition (1983), p. 212
Secondary source	Schmeh's Top 50 no. 28 (Cipherbrain, 2015); HCPortal transcription
Write-up	https://dbourdeau.github.io/cyphersolver/urquhart.html

Left out on purpose.

- **egmond**: independent re-solution; Lasry solved it in 2023 (key on cryptiana GL.htm)
- **lorraine1592**: independent re-solution; Tomokiyo solved it in 2025 (cryptiana unsolved list)
- **sp106r927**: independent re-solution; decoded by Sheila Richards, Secret Writings (1973), per Lasry
- **soria1523**: reclassified: key B was recovered from an external plaintext (the R9844 decipherment)

Primary source = the archive or printed original. Secondary source = where the item was found (DECODE, Tomokiyo's cryptiana, Cipherbrain, the BnF catalogue). Percentages are measured or estimated shares of the cipher text read, as recorded on each write-up.